

Grade 8 lessons

Exponents, roots, lines and systems

IN THIS BOOK

1. Exponents
2. Roots and right triangles
3. Linear equations
4. Slope and lines
5. Systems of equations

HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 8 Kahoot and Blooket sets, and one stop on the Grade 8 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Exponents

THE BIG IDEA

An exponent tells how many times to multiply a number by itself. 2^3 means $2 \times 2 \times 2$.

HOW TO DO IT

1. Same base, multiplying: add the exponents. $2^3 \times 2^4 = 2^7$.
2. Same base, dividing: subtract the exponents. $2^7 \div 2^3 = 2^4$.
3. Power of a power: multiply the exponents. $(5^2)^3 = 5^6$.
4. A negative exponent means "one over". $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$.
5. Scientific notation: 4.5×10^3 means move the decimal point 3 places right, giving 4500.

WORKED EXAMPLES

$$3^4 \times 3^5 = 3^{\square}$$

Same base, so add the exponents: $4 + 5 = 9$.

$$2^{-3} = ?$$

One over 2^3 . $2^3 = 8$, so the answer is $\frac{1}{8}$.

WATCH OUT

2^3 is 8, not 6. The exponent rules only work when the bases are the same.

WORDS TO KNOW

base

the number being multiplied

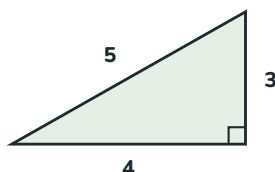
exponent

the small number that says how many times

Roots and right triangles

THE BIG IDEA

A square root undoes squaring. In any right triangle the squares of the two short sides add up to the square of the longest side.



A right triangle with legs 3 and 4. The longest side is 5.

HOW TO DO IT

1. $\sqrt{49} = 7$ because $7 \times 7 = 49$.
2. Pythagorean theorem: $a^2 + b^2 = c^2$, where c is the longest side.
3. To find c : square both legs, add them, then take the square root.
4. To find a leg: square the longest side, subtract the square of the other leg, then take the square root.

WORKED EXAMPLES

The legs are 6 and 8. How long is the longest side?

$$6^2 + 8^2 = 36 + 64 = 100. \sqrt{100} = 10.$$

The longest side is 13 and one leg is 5. Find the other leg.

$$13^2 - 5^2 = 169 - 25 = 144. \sqrt{144} = 12.$$

WATCH OUT

c is always the side across from the right angle. Add the squares before taking the root: $\sqrt{9 + 16}$ is 5, not $3 + 4$.

WORDS TO KNOW

hypotenuse

the longest side of a right triangle

leg

one of the two shorter sides

Linear equations

THE BIG IDEA

When x appears on both sides of an equation, collect the x terms on one side and the plain numbers on the other.

HOW TO DO IT

1. Expand any brackets first.
2. Add or subtract to move all the x terms to one side.
3. Add or subtract to move the plain numbers to the other side.
4. Divide to get x on its own.
5. If the x terms cancel completely, there is either no solution or every number works.

WORKED EXAMPLES

Solve $5x - 3 = 2x + 9$.

Subtract $2x$: $3x - 3 = 9$. Add 3: $3x = 12$. Divide by 3: $x = 4$.

Solve $3(x - 2) = 2x + 5$.

Expand: $3x - 6 = 2x + 5$. Subtract $2x$: $x - 6 = 5$. Add 6: $x = 11$.

WATCH OUT

When you expand a bracket, multiply every term inside. $3(x - 2)$ is $3x - 6$, not $3x - 2$.

WORDS TO KNOW

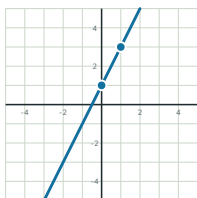
like terms

terms with the same letter and power

Slope and lines

THE BIG IDEA

Slope measures how steep a line is. It tells how far the line goes up for each step to the right.



From one dot to the next: right 1, up 2. The slope is 2.

HOW TO DO IT

1. Slope = rise $\div$ run = (change in y) $\div$ (change in x).
2. Pick two points. Subtract the y values. Subtract the x values in the same order. Divide.
3. In $y = mx + b$, m is the slope and b is where the line crosses the y -axis.
4. A positive slope goes uphill left to right. A negative slope goes downhill.

WORKED EXAMPLES

Find the slope through (1, 2) and (3, 8).

Rise: $8 - 2 = 6$. Run: $3 - 1 = 2$. Slope: $6 \div 2 = 3$.

For $y = 3x - 2$, what are the slope and y -intercept?

The slope is 3. The line crosses the y -axis at -2 .

WATCH OUT

Rise goes on top, run on the bottom. Subtract in the same order for both, or the sign will be wrong.

WORDS TO KNOW

slope

the steepness of a line

y -intercept

where the line crosses the y -axis

Systems of equations

THE BIG IDEA

A system is two equations with the same x and y . The solution is the one point that makes both equations true.

HOW TO DO IT

1. If both equations start with " $y =$ ", set the two right sides equal and solve for x .
2. Put that x back into either equation to find y .
3. If the equations are lined up (like $x + y$ and $x - y$), add or subtract them to remove one letter.
4. Write the answer as a point (x, y) and check it in both equations.

WORKED EXAMPLES

Solve $y = 2x + 1$ and $y = x + 4$.

Set them equal: $2x + 1 = x + 4$, so $x = 3$. Then $y = 3 + 4 = 7$. The solution is $(3, 7)$.

Solve $x + y = 10$ and $x - y = 4$.

Add the equations: $2x = 14$, so $x = 7$.
Then $7 + y = 10$, so $y = 3$.

WATCH OUT

Find both x and y . Then check the point in both equations, not just one.

WORDS TO KNOW

system

two or more equations solved together

substitute

replace a letter with what it equals